EE 121B: Principles of Semiconductor Device Design

Lecture 3:

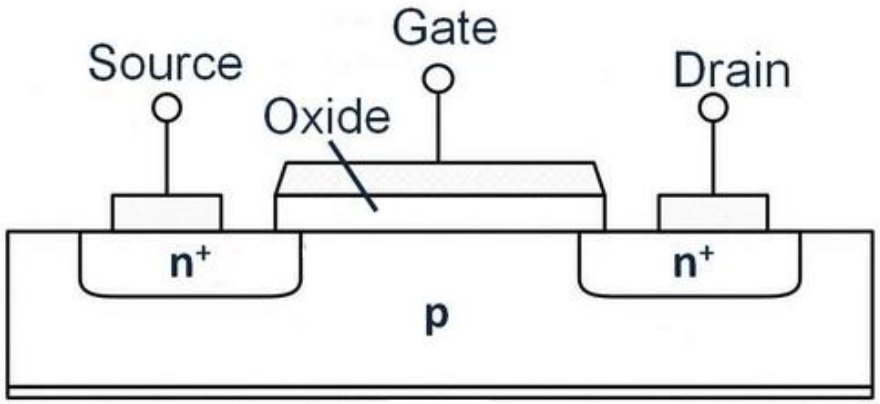
Semiconductor Fundamentals (continued)

Sam Emaminejad

Course Overview

1. Semiconductor Properties
2. P-N Junction
3. Metal-Semiconductor Contacts
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



http://qph.is.quoracdn.net/main-qimg-38a99b4f215e161bf31964795cfce7d2?convert_to_webp=true

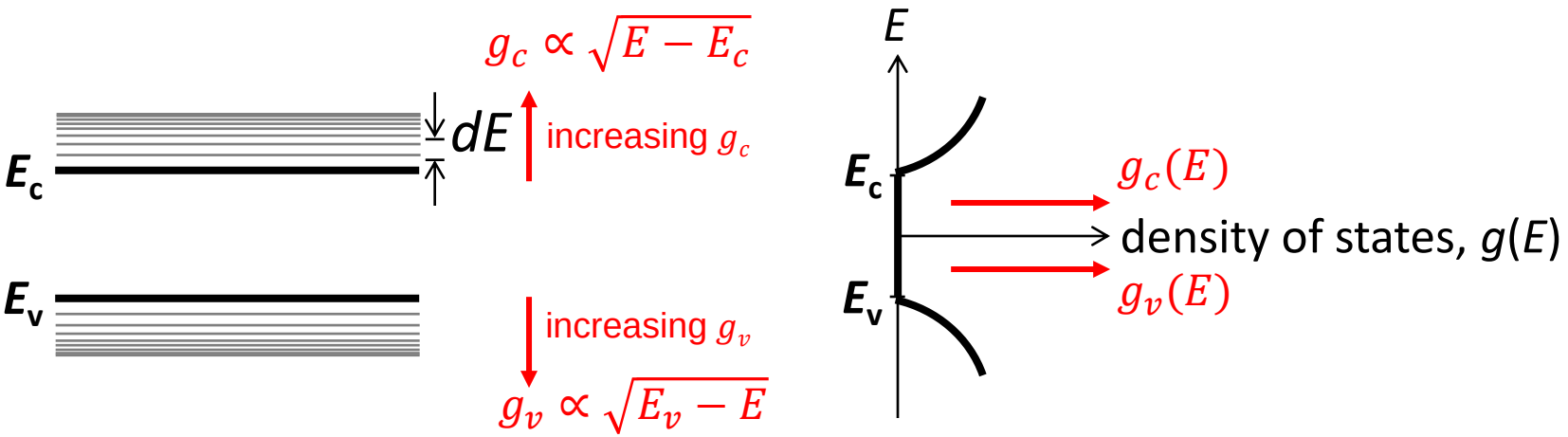
Lecture 3

OUTLINE

- Semiconductor Fundamentals (cont'd)
  - Relationship between E_F and n, p
  - Degenerately doped semiconductor
  - Properties of carriers in semiconductors
  - Carrier drift
    - Scattering mechanisms
    - Drift current
  - Conductivity and resistivity

Reading: Pierret 2.4-2.5, 3.1; Hu 1.5, 1.7-1.10, 2.1-2.2

Recall: Density of States



g*(*E*)*dE* = number of states per cm³ in the energy range between *E* and *E*+*dE

Near the band edges:

$$g_c(E) = \frac{8\pi\sqrt{2}}{h^3}\left(m^*_{n,DOS}\right)^{3/2}\sqrt{(E - E_c)} \quad \text{for } E \geq E_c$$

$$g_v(E) = \frac{8\pi\sqrt{2}}{h^3}\left(m^*_{p,DOS}\right)^{3/2}\sqrt{(E_v - E)} \quad \text{for } E \leq E_v$$

Electron and hole density-of-states effective masses

	Si	**Ge**	**GaAs**
$m_{n,DOS}^*/m_o$	**1.08**	0.56	0.067
$m_{p,DOS}^*/m_o$	**0.81**	0.29	0.47

$m_0 = 9.1 \times 10^{-31}$ kg

Boltzmann Approximation

- Probability that an available state at energy *E* is occupied:

Fermi Function: $$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

- E_F is called the ***Fermi energy*** or the ***Fermi level***

$$\text{If } E - E_F > 3kT,\ f(E) \cong e^{-(E - E_F)/kT}$$

$e^{(E-E_F)/kT} \gg 1 \rightarrow f(E) = \frac{1}{e^{(E-E_F)/kT}}$

$$\text{If } E_F - E > 3kT,\ f(E) \cong 1 - e^{(E - E_F)/kT}$$

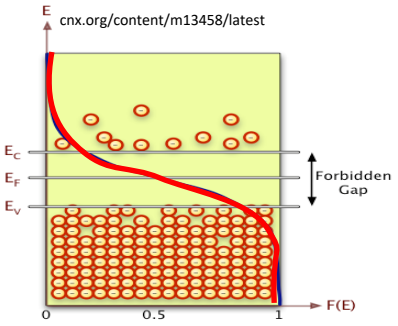
If $|x| \ll 1$ and $|\alpha x| \ll 1$ $\quad (1 + x)^{\alpha} \approx 1 + \alpha x, \alpha = -1, x = e^{(E-E_F)/kT}$

Probability that a state is empty (*i.e.* occupied by a hole):

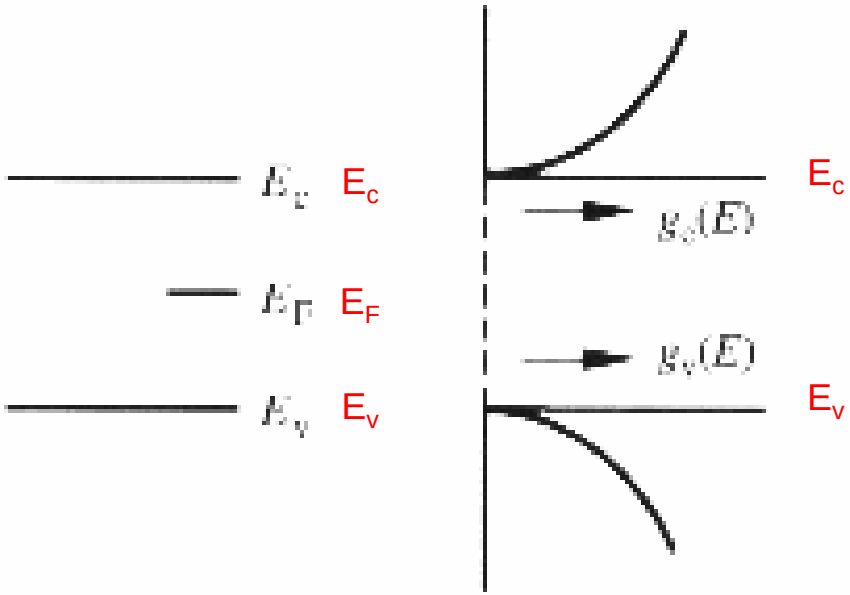
$$1 - f(E) \cong e^{(E - E_F)/kT} = e^{-(E_F - E)/kT}$$

Equilibrium Distribution of Carriers

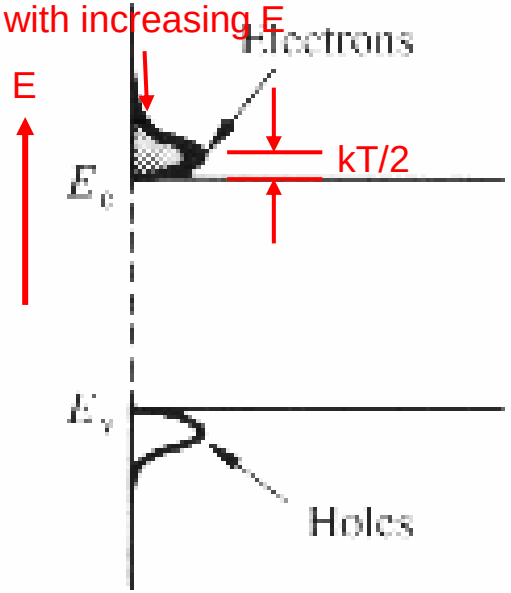
- Obtain $n(E)$ by multiplying $g_c(E)$ and $f(E)$



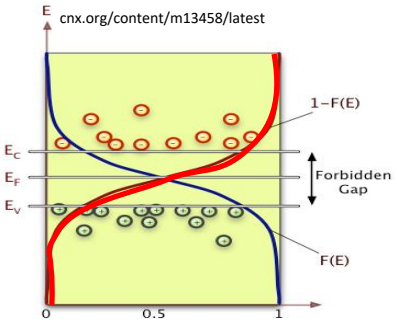
Energy band diagram | **Density of States, $g_c(E)$** | × | **Probability of occupancy, $f(E)$** | = | **Carrier distribution, $n(E)$**



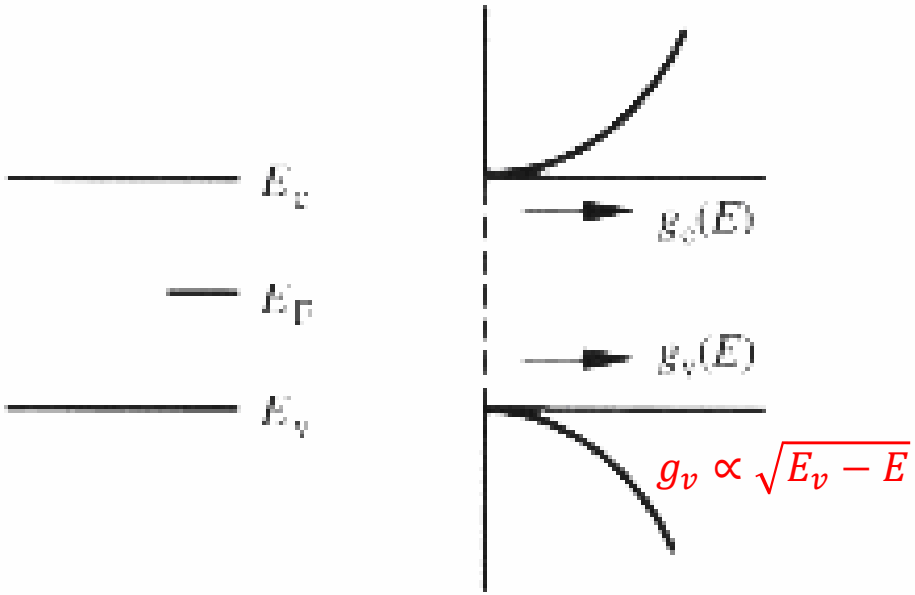
Decrease exponentially with increasing E



- Obtain $p(E)$ by multiplying $g_v(E)$ and 1-$f(E)$



Energy band diagram | **Density of States, $g_v(E)$** × **Probability of occupancy, 1-$f(E)$** = **Carrier distribution, $p(E)$**



Equilibrium Carrier Concentrations

- Integrate $n(E)$ over all the energies in the conduction band to obtain n:

$$n = \int_{E_c}^{\infty} g_c(E) f(E) dE$$

(∞: top of conduction band; $f(E) \sim e^{-(E-E_F)/kT}$)

- <u>By using the Boltzmann approximation</u>, and extending the integration limit to ∞, we obtain

$$n = N_c e^{-(E_c - E_F)/kT} \quad \text{where} \quad N_c = 2\left(\frac{2\pi m^*_{n,DOS} kT}{h^2}\right)^{3/2}$$

N_c is the "effective" densities of states @ the conduction band edge

$= 2.82 \times 10^{19}$ cm^{-3} in Si @ 300 K

- Integrate $p(E)$ over all the energies in the valence band to obtain p:

$$p = \int_{\text{bottom of valence band}}^{E_v} g_v(E)\underbrace{[1 - f(E)]}_{\sim e^{-(E_F - E)/kT}} dE$$

$-\infty$

- By using the Boltzmann approximation, and extending the integration limit to -∞, we obtain

$$p = N_v e^{-(E_F - E_v)/kT} \quad \text{where} \quad N_v = 2\left(\frac{2\pi m^*_{p,DOS} kT}{h^2}\right)^{3/2}$$

N_v is the "effective" densities of states @ the valence band edge

$= 1.83 \times 10^{19}$ cm^{-3} in Si @ 300 K

Intrinsic Carrier Concentration

$$np = \left(N_c e^{-(E_c - E_F)/kT}\right)\left(N_v e^{-(E_F - E_v)/kT}\right)$$

$$= N_c N_v e^{-(E_c - E_v)/kT} = N_c N_v e^{-E_G/kT} = n_i^2$$

$$n_i = \sqrt{N_c N_v}\, e^{-E_G/2kT}$$

Effective Densities of States at the Band Edges (@ 300K)

	Si	Ge	GaAs
N_c (cm^{-3})	**2.82×10^{19}**	1.05×10^{19}	4.37×10^{17}
N_v (cm^{-3})	**1.83×10^{19}**	3.92×10^{18}	8.12×10^{18}

n(n_i, E_i) and p(n_i, E_i)

- **In an intrinsic semiconductor, $n = p = n_i$ and $E_F = E_i$**

$$n = n_i = N_c e^{-(E_c - E_i)/kT}$$

$$\Rightarrow N_c = n_i e^{(E_c - E_i)/kT}$$

$$n = \underbrace{[n_i e^{(E_c - E_i)/kT}]}_{N_c} e^{-(E_c - E_F)/kT}$$

$$p = n_i = N_v e^{-(E_i - E_v)/kT}$$

$$\Rightarrow N_v = n_i e^{(E_i - E_v)/kT}$$

$$p = \underbrace{[n_i e^{(E_i - E_v)/kT}]}_{N_v} e^{-(E_F - E_v)/kT}$$

$$\boxed{n = n_i e^{(E_F - E_i)/kT}}$$

$$\boxed{p = n_i e^{(E_i - E_F)/kT}}$$

Intrinsic Fermi Level, E_i

- To find E_F for an intrinsic semiconductor, use the fact that $n = p$:

$$N_c e^{-(E_c - E_F)/kT} = N_v e^{-(E_F - E_v)/kT}$$

$$\rightarrow \frac{N_v}{N_c} = e^{-\frac{E_c - E_F}{kT}} / e^{-\frac{E_F - E_v}{kT}}$$

$$\rightarrow \mathrm{kTln}\left(\frac{N_v}{N_c}\right) = -(E_c - E_F) + (E_F - E_v)$$

$$= 2E_F - (E_c + E_v)$$

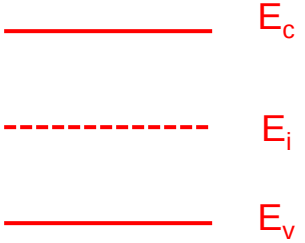
$$E_F = \frac{E_c + E_v}{2} + \frac{kT}{2}\ln\left(\frac{N_v}{N_c}\right) \equiv E_i$$

$$E_i = \frac{E_c + E_v}{2} + \frac{3kT}{4}\ln\left(\frac{m^*_{p,DOS}}{m^*_{n,DOS}}\right) \cong \frac{E_c + E_v}{2}$$

0.81 m_0

1.08 m_0

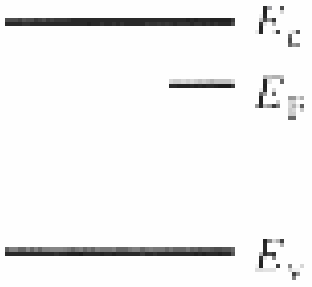
-0.006 eV



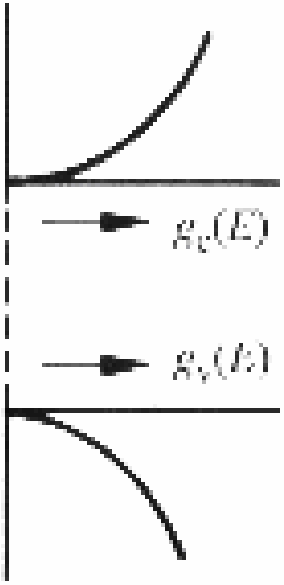
n-type Material

R. F. Pierret, *Semiconductor Device Fundamentals*, Figure 2.16

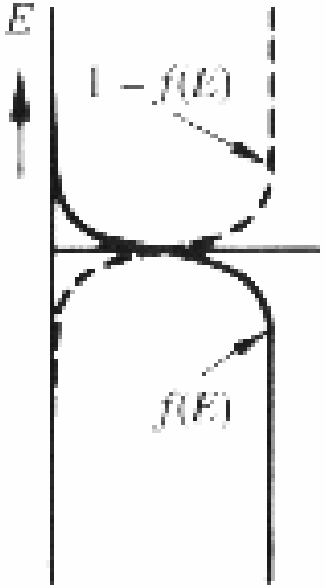
Energy band diagram



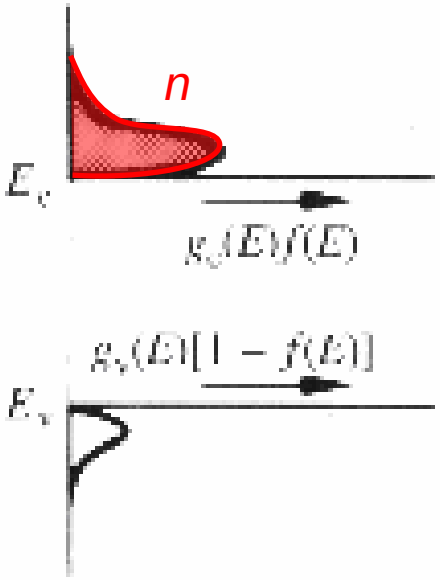
Density of States



Probability of occupancy



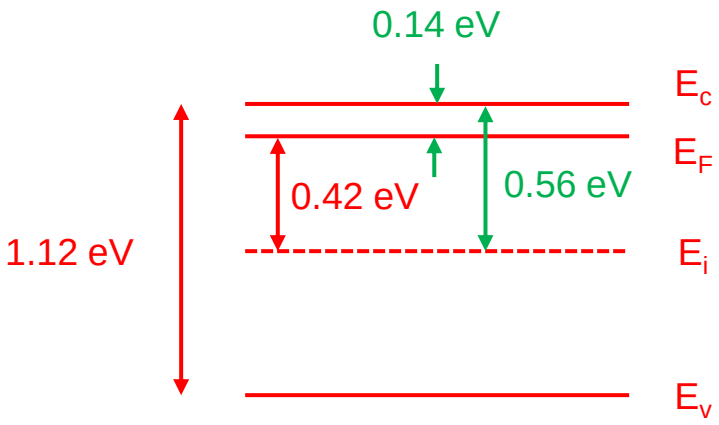
Carrier distributions



Fermi level is closer to E_c than E_v, i.e., $E_f > E_i$

Example: Energy-band diagram

Question: Where is E_F for $n = 10^{17}$ cm^{-3} (at 300 K) ?



$$n = n_i e^{(E_i - E_F)/kT} = 10^3$$

$$E_F - E_i = 0.14 \text{ eV}$$

$$n = n_i e^{(E_F - E_i)/kT} = 10^{17}$$

$$\uparrow$$

$$10^{10}$$

$$\ln(10^7) = (E_F - E_i)/kT$$

$$E_F - E_i = kT\ln(10^7)$$

$$= 7 \times kT\ln(10)$$

$$= 7 \times 0.060\ eV$$

$$= 0.42\ eV$$

Example: Dopant Ionization

Consider a phosphorus-doped Si sample at 300K with $N_D = 10^{17}$ cm^{-3}.

What fraction of the donors are not ionized?

Hint: Suppose at first that **all** of the donor atoms are ionized.

First, assume: $n = ND$ (i.e., 100% ionization)

Recall:

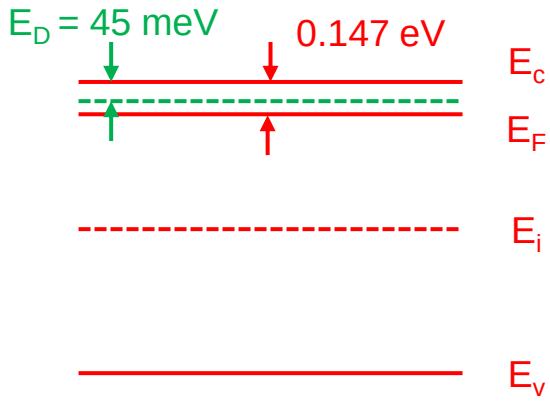
$n = N_c e^{-(E_c - E_F)/kT}$

$$E_F = E_c - kT \ln\left(\frac{N_c}{n}\right) = E_c - 147meV$$

Probability of non-ionization $\cong \dfrac{1}{1+e^{(E_D - E_F)/kT}}$

Recall:

$f(E) = \dfrac{1}{1+e^{(E-E_F)/kT}}$



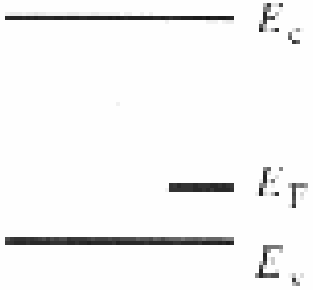
$$= \frac{1}{1+e^{(147meV - 45meV)/26meV}} = 0.02$$

$n \cong 9.8 \times 10^{16}$ cm^{-3}

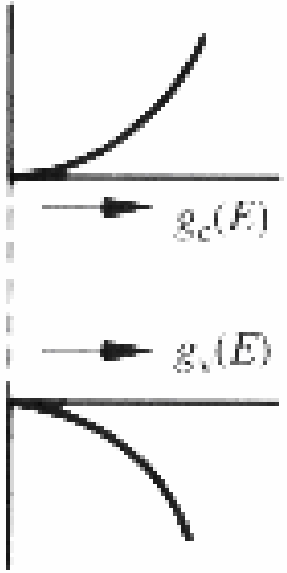
p-type Material

R. F. Pierret, *Semiconductor Device Fundamentals*, Figure 2.16

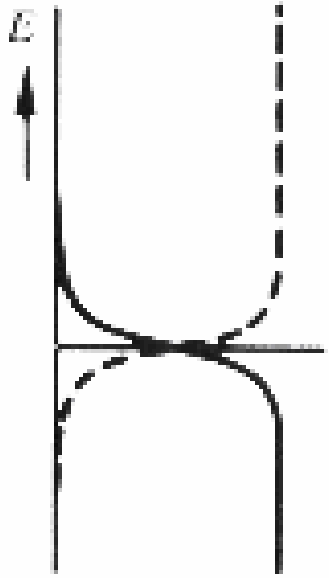
Energy band diagram



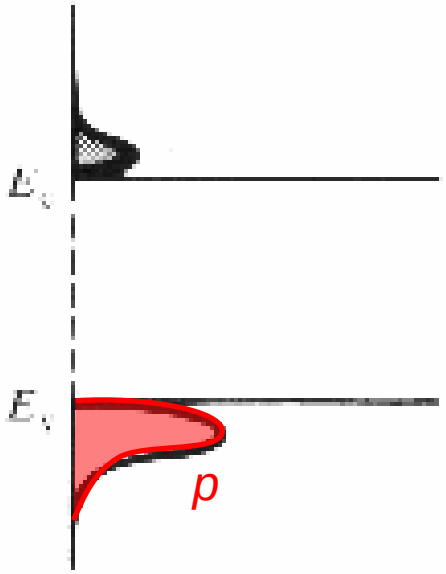
Density of States



Probability of occupancy



Carrier distributions

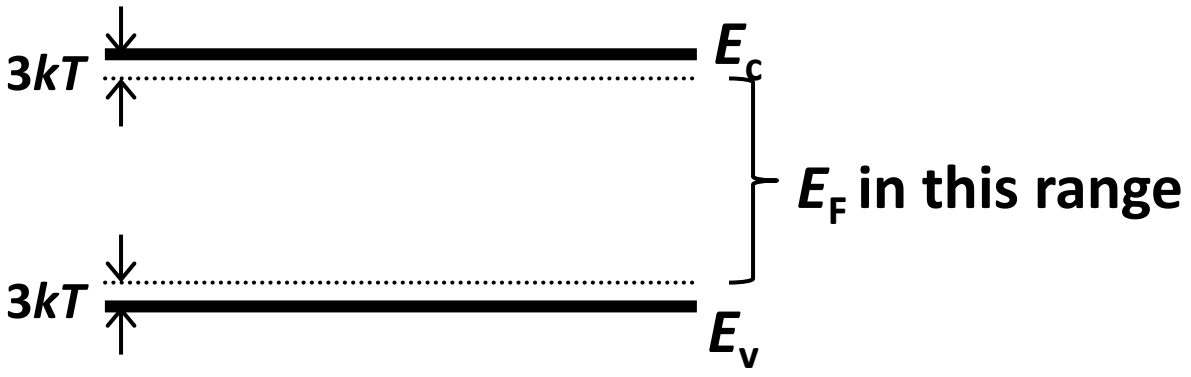


Fermi level is closer to E_v than E_c, i.e., $E_f < E_i$

Non-degenerately Doped Semiconductor

- Recall that the expressions for *n* and *p* were derived using the Boltzmann approximation, *i.e.* we assumed

$$E_v + 3kT \leq E_F \leq E_c - 3kT$$



The semiconductor is said to be ***non-degenerately doped*** in this case.

Degenerately Doped Semiconductor

- If a semiconductor is very heavily doped, the Boltzmann approximation is not valid.

In Si at T=300K: $E_c - E_F < 3k_BT$ if $N_D > 1.6\text{x}10^{18}$ cm^{-3}

~10^{18} cm^{-3}

$E_F - E_v < 3k_BT$ if $N_A > 9.1\text{x}10^{17}$ cm^{-3}

The semiconductor is said to be ***degenerately doped*** in this case.

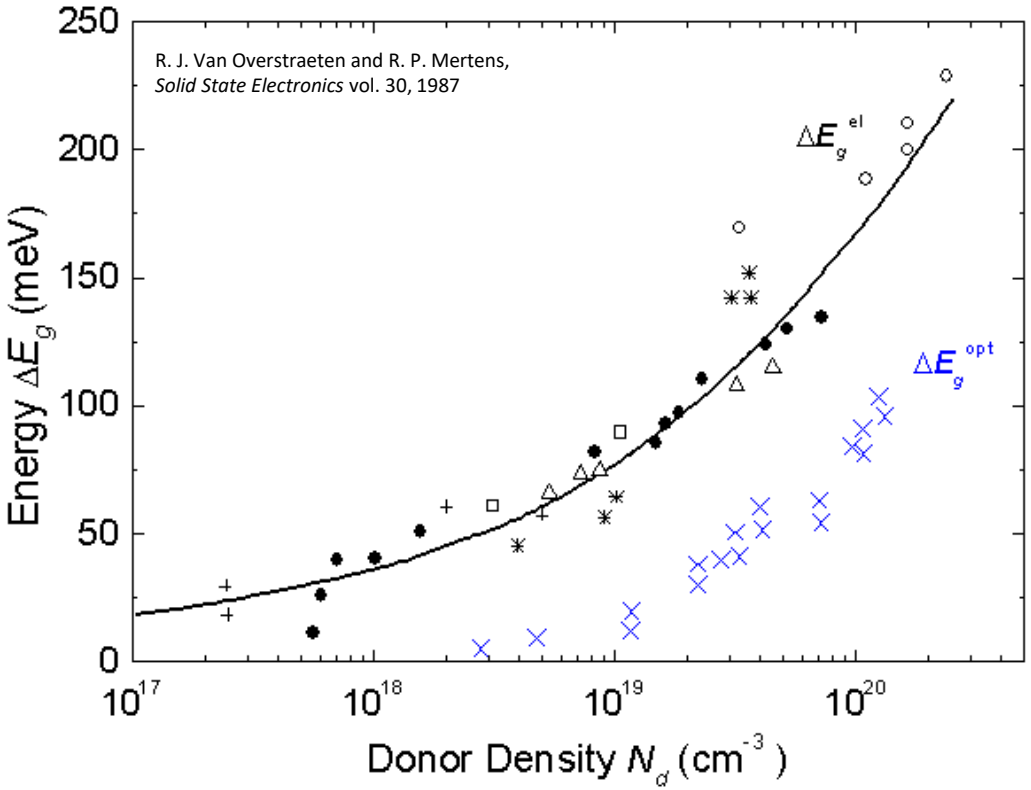
- Terminology:

"n+" → degenerately n-type doped. $E_F \cong E_c$

"p+" → degenerately p-type doped. $E_F \cong E_v$

Band Gap Narrowing

- If the dopant concentration is a significant fraction of the silicon atomic density, the energy-band structure is perturbed → the band gap is reduced by ΔE_G :



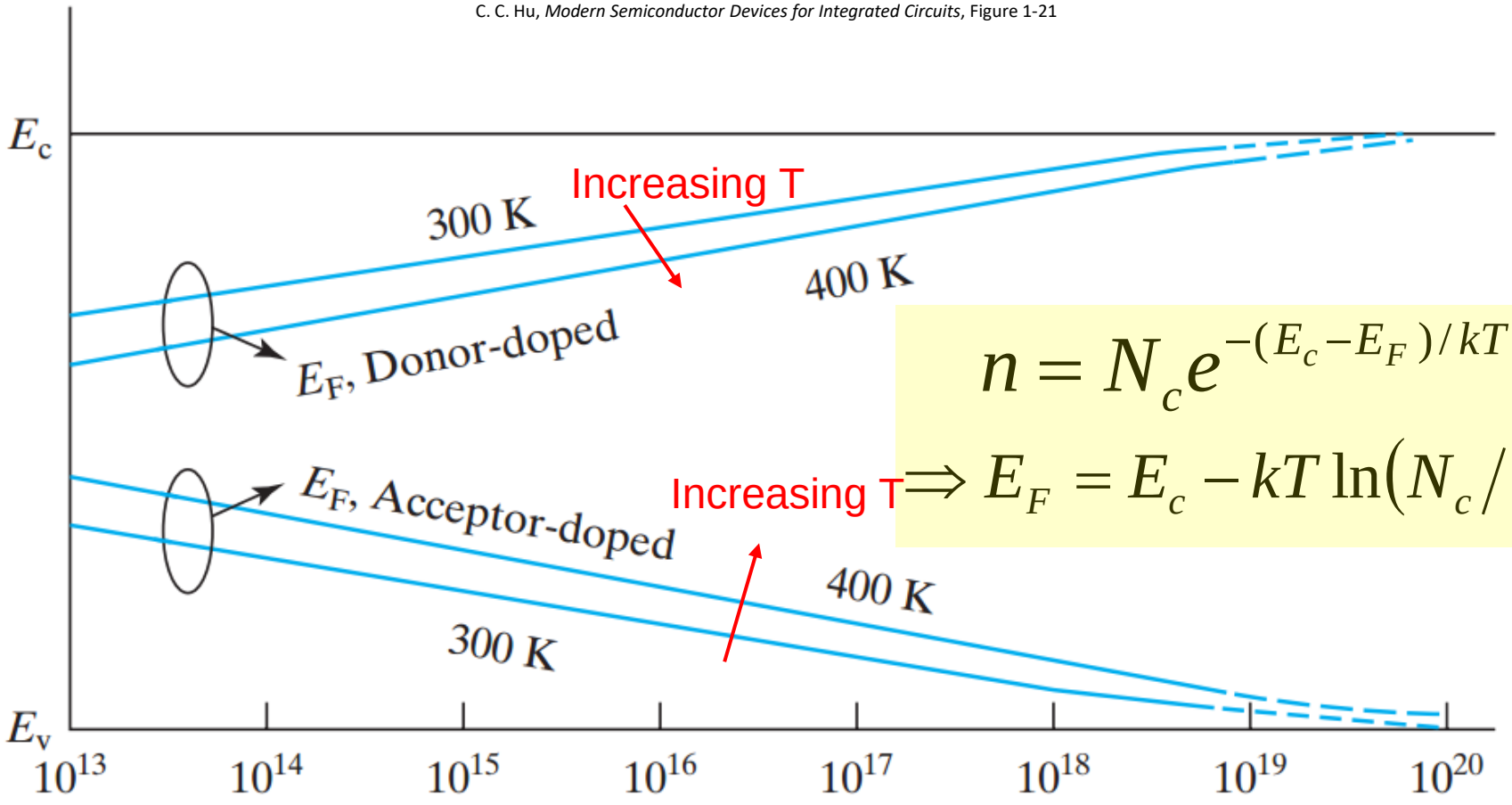
N ≥ 10^{18} cm^{-3}

$$\Delta E_G \cong 3.5\times10^{-8} N^{1/3} \sqrt{\frac{300}{T}}$$

$N = 10^{18}$ cm^{-3}: ΔE_G = 35 meV

$N = 10^{19}$ cm^{-3}: ΔE_G = 75 meV

Dependence of E_F on Temperature



$$n = N_c e^{-(E_c - E_F)/kT}$$

$$\Rightarrow E_F = E_c - kT \ln(N_c/n)$$

Mobile Charge Carriers in Semiconductors

- Three primary types of carrier action occur inside a semiconductor:
  - **Drift:** charged particle motion under the influence of an electric field.

Current flow in a MOSFET

  - **Diffusion:** particle motion due to concentration gradient or temperature gradient.

Current flow in a PN diode, and a BJT

  - **Recombination-generation** (R-G)

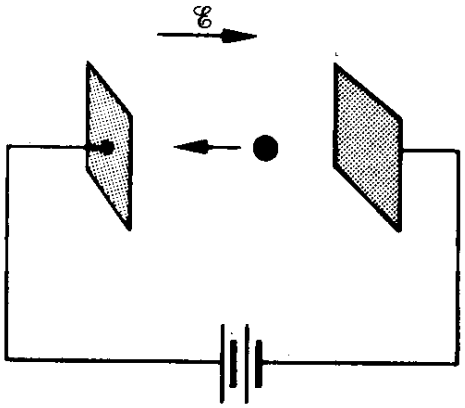
Leveraged in light-emitting diodes

Leveraged in solar cells

Electrons as Moving Particles

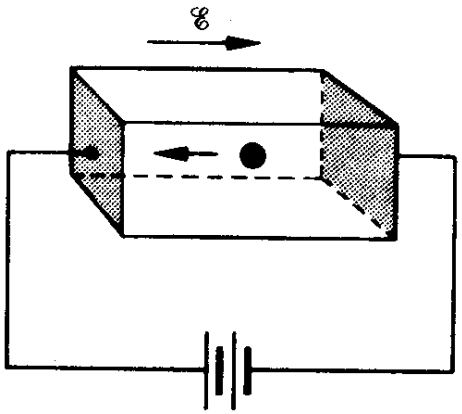
R.F. Pierret, *Semiconductor Fundamentals*, Figure 2.9

In vacuum



$F = (-q)\mathcal{E} = m_o a$

In semiconductor



$F = (-q)\mathcal{E} = m_n^* a$

where m_n^* is the ***conductivity effective mass***

Conductivity Effective Mass, *m**

Under the influence of an electric field (E-field), an electron or a hole is accelerated:

$$a = \frac{-q\mathcal{E}}{m_n^*} \quad \text{electrons}$$

$$a = \frac{q\mathcal{E}}{m_p^*} \quad \text{holes}$$

Electron and hole conductivity effective masses

	Si	**Ge**	**GaAs**
m_n^*/m_o	**0.26**	0.12	0.068
m_p^*/m_o	**0.39**	0.30	0.50

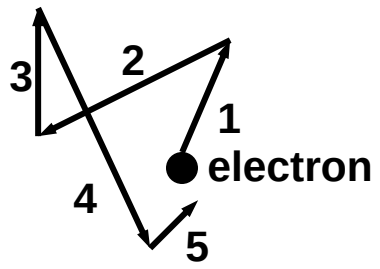
$m_o = 9.1\times10^{-31}$ kg

Carrier Scattering

- Mobile electrons and atoms in the Si lattice are always in random thermal motion.
  - Electrons make frequent collisions with the vibrating atoms

    "lattice scattering" or **"phonon scattering"** – increases with increasing T
- Other scattering mechanisms:
  - deflection by ionized impurity atoms
  - deflection due to Coulombic force between carriers

    "carrier-carrier scattering" – only significant at high carrier concentrations
- **The net current in any direction is zero, if no E-field is applied.**



Thermal Velocity, v_{th}

Average electron kinetic energy $= \frac{3}{2}kT = \frac{1}{2}m_n^* v_{th}^2$

$$v_{th} = \sqrt{\frac{3kT}{m_n^*}} = \sqrt{\frac{3 \times 0.026\text{eV} \times (1.6 \times 10^{-19}\,\text{J/eV})}{0.26 \times 9.1 \times 10^{-31}\,\text{kg}}}$$

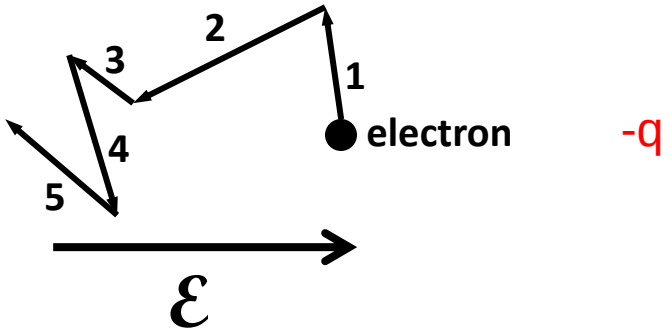
$$= 2.3 \times 10^5\,\text{m/s} = 2.3 \times 10^7\,\text{cm/s}$$ For electrons

For holes, $m_p^* = 0.39\ m_0$, so

$$v_{th,p} = 1.9 \times 10^7\ cm/s$$

Carrier Drift

- When an electric field (*e.g.* due to an externally applied voltage) exists within a semiconductor, mobile charge-carriers will be accelerated by the electrostatic force:



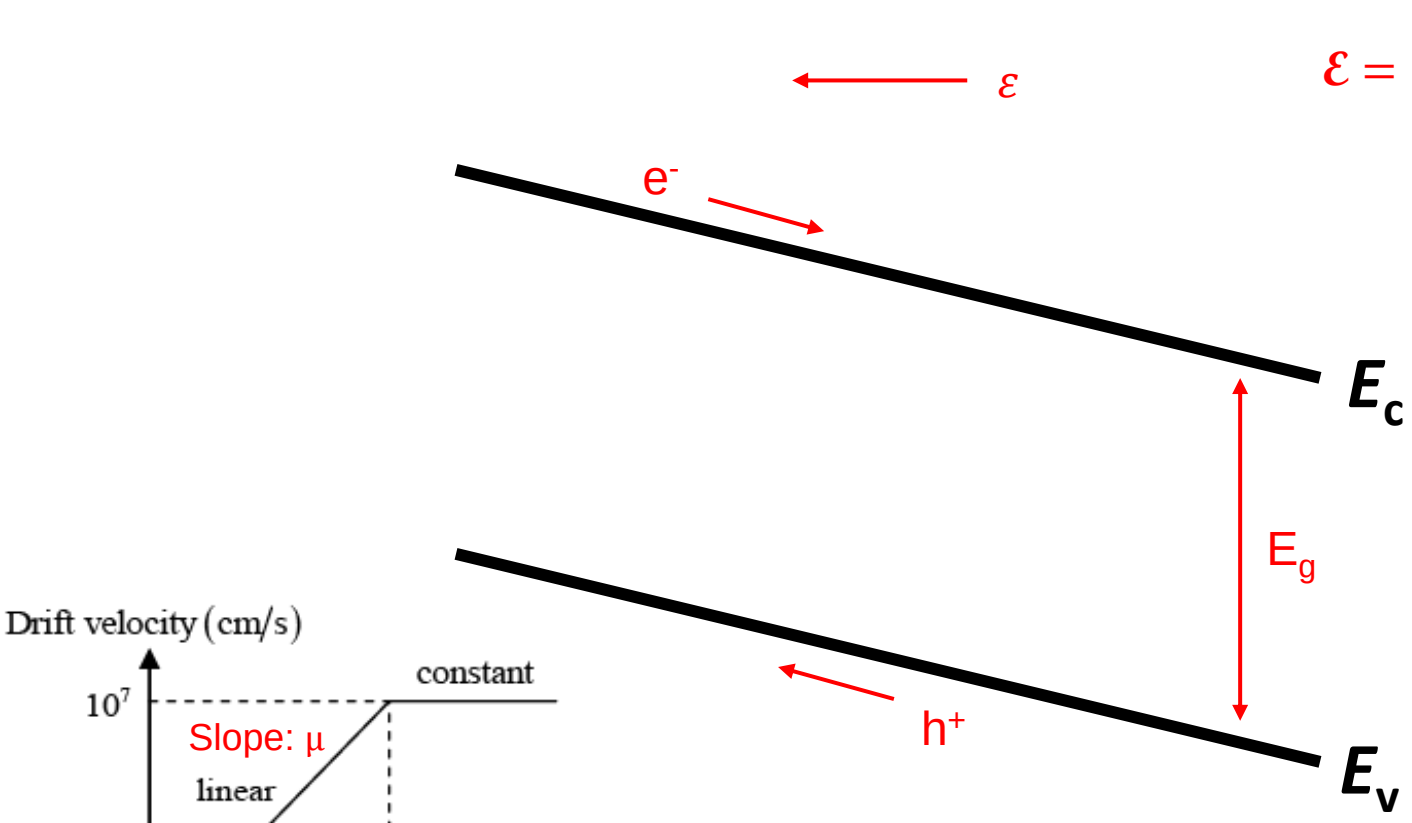
Electrons *drift* in the direction opposite to the E-field → net current

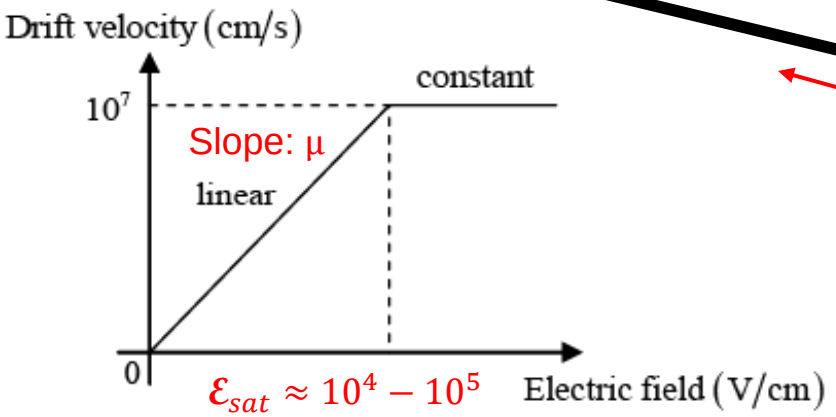
Because of scattering, electrons in a semiconductor do not undergo constant acceleration. **However, they can be viewed as quasi-classical particles moving at a constant average *drift velocity* v_{dn}**

$$v_{dn} = \mu_n \mathcal{E}$$

Carrier Drift (Band Model)

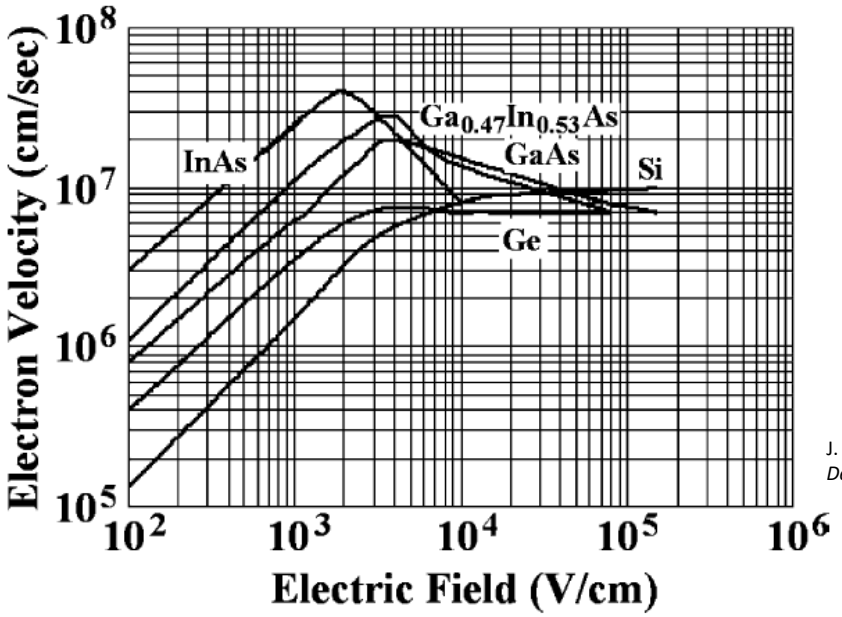
$$\mathcal{E} = \frac{1}{q}\frac{dE_c}{dx} = \frac{1}{q}\frac{dE_v}{dx}$$





Velocity Saturation

- **At high electric field, carrier drift velocity saturates:**



J. Bean, in *High-Speed Semiconductor Devices*, S.M. Sze (ed.), 1990

The *saturation velocity,* v_{sat}, is the maximum drift velocity

$$v_{sat} = \begin{cases} 8\times10^6 \text{ cm/s} & \text{for electrons in Si} \\ 6\times10^6 \text{ cm/s} & \text{for holes in Si} \end{cases}$$

Electron Momentum

- With every collision, the electron loses momentum

$$m_n^* v_{dn}$$

- Between collisions, the electron gains momentum

$$-q\mathcal{E}\tau_{mn}$$

τ_{mn} ≡ average time between electron scattering events

Conservation of momentum → $|m_n^* v_{dn}| = |q\mathcal{E}\tau_{mn}|$

Carrier Mobility, μ

For electrons: $|\boldsymbol{v}_{\mathbf{dn}}| = q\mathcal{E}\tau_{mn} / m_n^* \equiv \mu_n\mathcal{E}$

$\mu_n \equiv [q\tau_{mn} / m_n^*]$ is the ***electron mobility***

Similarly, for holes: $|v_{dp}| = q\mathcal{E}\tau_{mp} / m_p^* \equiv \mu_p\mathcal{E}$

$\mu_p \equiv [q\tau_{mp} / m_p^*]$ is the ***hole mobility***

Electron and hole mobilities for intrinsic semiconductors @ 300K

	Si	**Ge**	**GaAs**	**InAs**
μ_n ($cm^2/V\cdot s$)	**1400**	3900	8500	30,000
μ_p ($cm^2/V\cdot s$)	**470**	1900	400	500

Example: Drift Velocity Calculation

a) Find the hole drift velocity in an intrinsic Si sample for $\mathcal{E}$ = 10^3 V/cm.

b) What is the average hole scattering time?

Solution:

a) $v_{dp} = \mu_p \mathcal{E}$ $= 470 \times 10^3 = 4.7 \times 10^5$ cm/s

cm^2/Vs V/cm

b) $\mu_p = \dfrac{q\tau_{mp}}{m_p^*} \Rightarrow \tau_{mp} = \dfrac{m_p^* \mu_p}{q}$ $= \dfrac{(0.39 \times 9.1 \times 10^{-31}\text{kg})(470\ \text{cm}^2/\text{V}\cdot\text{s})}{1.6 \times 10^{-19}\ \text{C}}$

$\sim 10^{-9} \dfrac{\text{kg cm}^2/\text{V}\cdot\text{s}}{\text{C}} = 10^{-13}\ \text{s} = 0.1\ \text{ps}$

Mean Free Path

- Average distance traveled between collisions

$$l = v_{th}\tau_{mp}$$

$$= 1.9 \times 10^{7}\ \text{cm/s} \times 10^{-13}\text{s}$$

$$= 19\ \text{nm}$$

Mechanisms of Carrier Scattering

Dominant scattering mechanisms:

1. Phonon scattering (lattice scattering)
2. Impurity (dopant) ion scattering

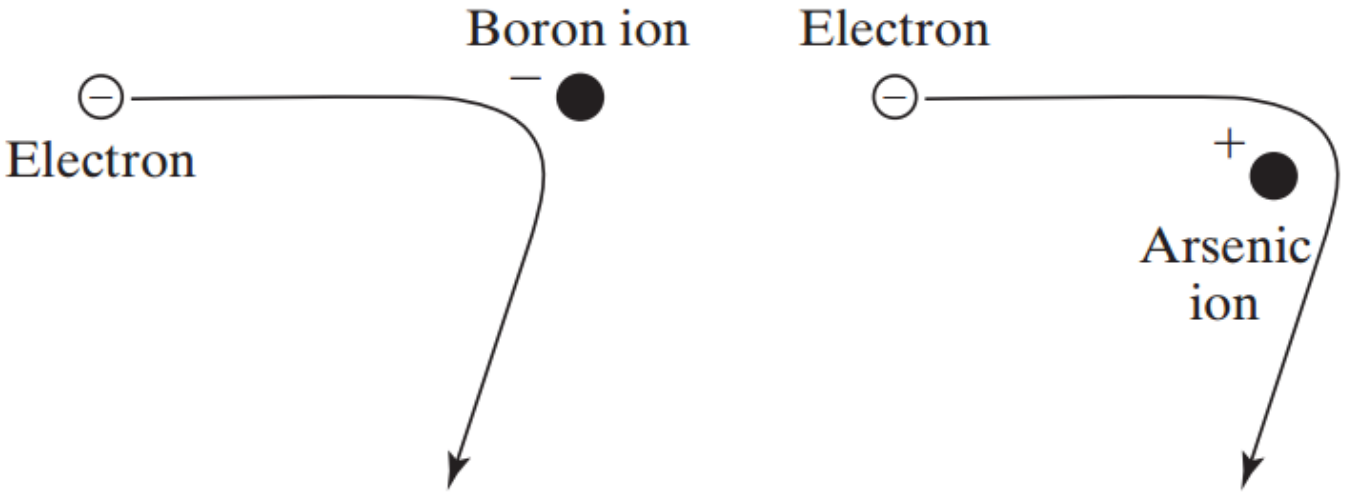
Phonon scattering limited mobility decreases with increasing *T*:

$$\mu_{phonon} \propto \tau_{phonon} \propto \frac{1}{\text{phonon density} \times \text{carrier thermal velocity}} \propto \frac{1}{T \times T^{1/2}} \propto T^{-3/2}$$

$\mu = q\tau / m$

$v_{th} \propto \sqrt{T}$

Impurity Ion Scattering



There is less change in the electron's direction if the electron travels by the ion at a higher speed.

Ion scattering limited mobility increases **with increasing *T*:**

$$\mu_{impurity} \propto \frac{v_{th}^3}{N_A + N_D} \propto \frac{T^{3/2}}{N_A + N_D}$$

Matthiessen's Rule

- The probability that a carrier will be scattered by mechanism *i* within a time period *dt* is dt/τ_i

τ_i ≡ mean time between scattering events due to mechanism *i*

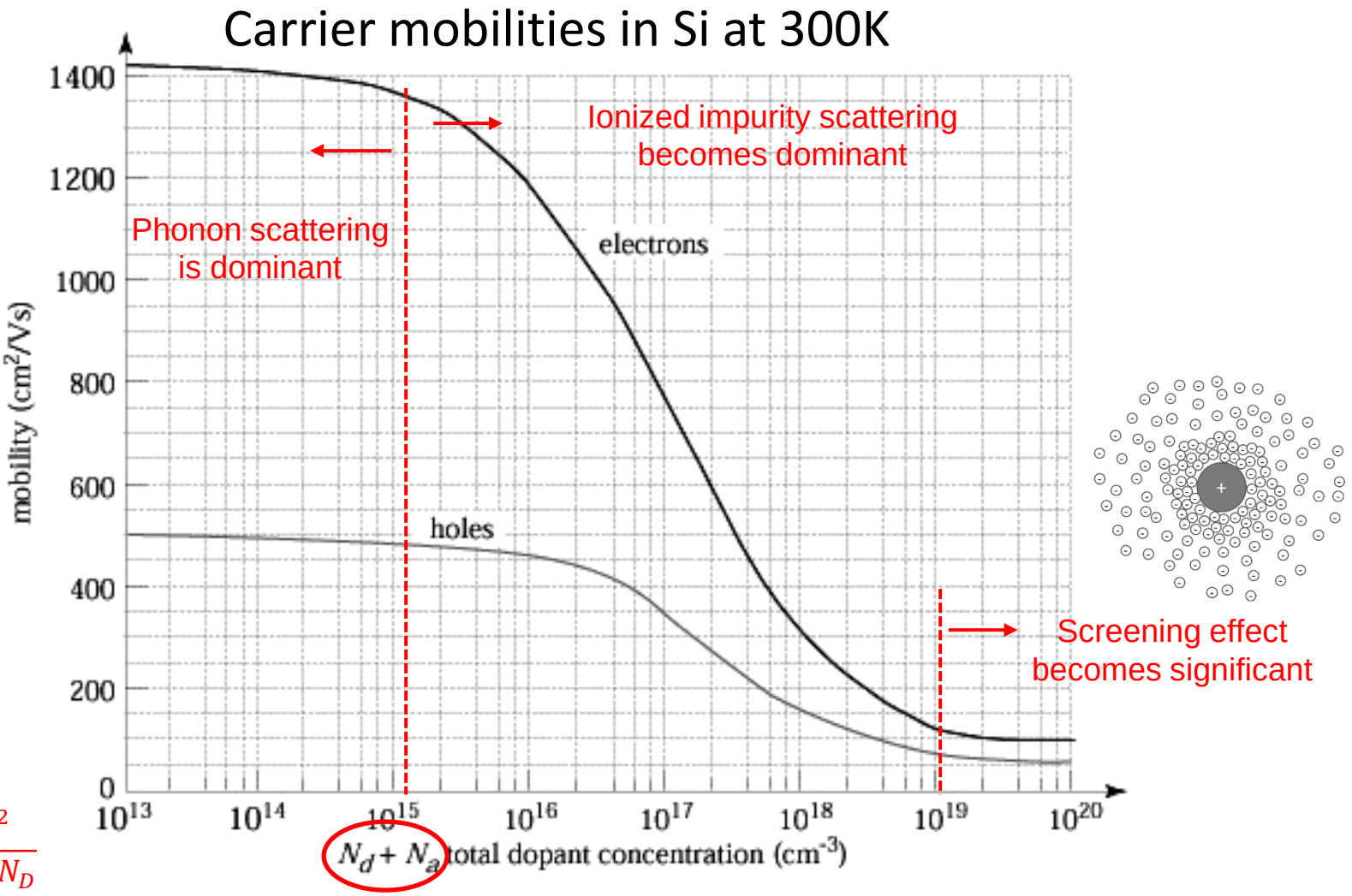
→ Probability that a carrier will be scattered by any mechanism within a time period *dt* is $\sum_i dt/\tau_i$

$\mu = \frac{q\tau}{m^*}$

$$\frac{1}{\tau} = \frac{1}{\tau_{phonon}} + \frac{1}{\tau_{impurity}} \quad \Rightarrow \quad \frac{1}{\mu} = \frac{1}{\mu_{phonon}} + \frac{1}{\mu_{impurity}}$$

The dominant scattering mechanism (with the lowest associated μ_i) determines μ

Mobility Dependence on Doping

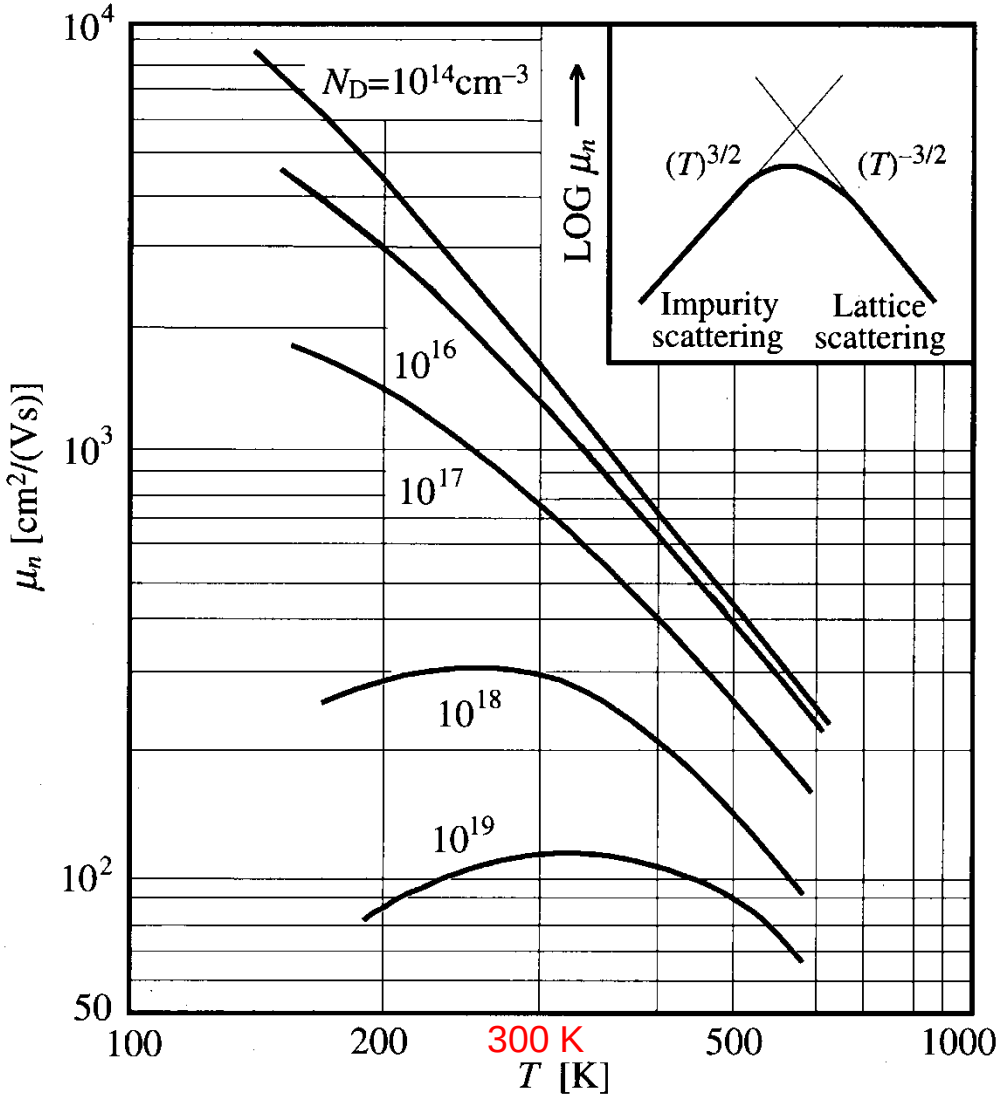


$$\mu_{impurity} \propto \frac{T^{3/2}}{N_A + N_D}$$

$$\mu_{phonon} \propto T^{-3/2}$$

http://large.stanford.edu/courses/2007/ap272/peng1/

Mobility Dependence on Temperature

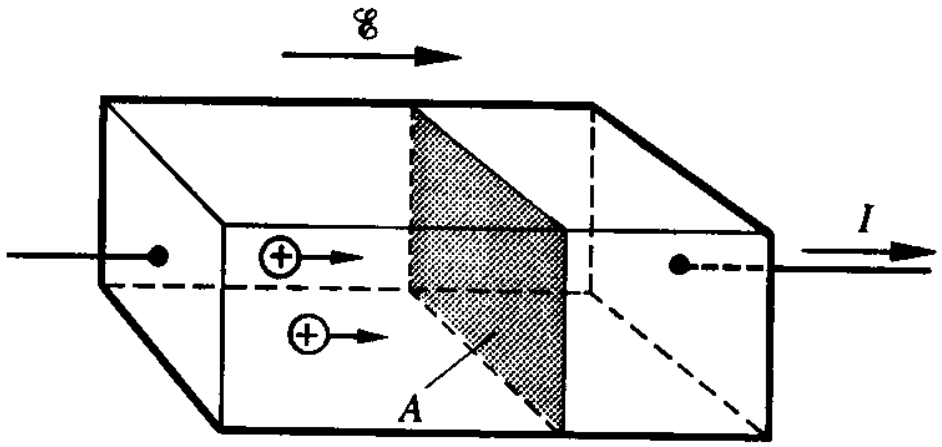


$$\frac{1}{\mu} = \frac{1}{\mu_{phonon}} + \frac{1}{\mu_{impurity}}$$

$\mu_{phonon} \propto T^{-3/2}$

$\mu_{impurity} \propto \frac{T^{3/2}}{N_A + N_D}$

Hole Drift Current Density, $J_{p,drift}$



R.F. Pierret, *Semiconductor Fundamentals*, Figure 3.3

$v_{dp}\,\Delta t\,A$ = volume from which all holes cross plane in time Δt

$p\,v_{dp}\,\Delta t$ A = number of holes crossing plane in time Δt

$q\,p\,v_{dp}\,\Delta t$ A = hole charge crossing plane in time Δt

$q\,p\,v_{dp}\,A$ = hole charge crossing plane per unit time = hole current

→ **Hole drift current per unit area $J_{p,drift} = q\,p\,v_{dp}$**

Conductivity and Resistivity

- In a semiconductor, both electrons and holes conduct current:

$$J_{p,drift} = qp\mu_p \mathcal{E} \qquad J_{n,drift} = -qn(-\mu_n \mathcal{E})$$

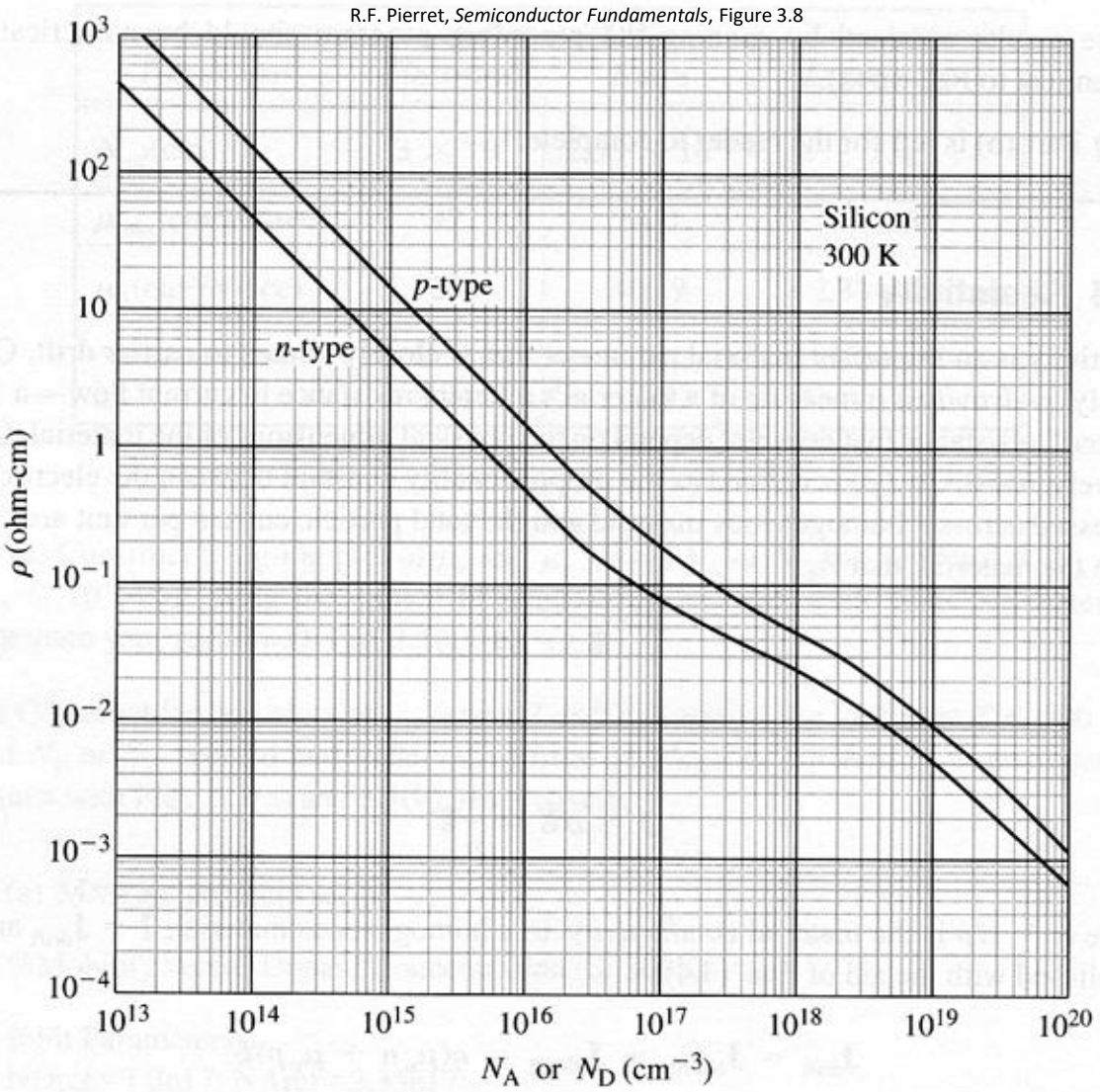
$$J_{drift} = J_{p,drift} + J_{n,drift} = qp\mu_p \mathcal{E} + qn\mu_n \mathcal{E}$$

$$J_{drift} = (qp\mu_p + qn\mu_n)\mathcal{E} \equiv \sigma \mathcal{E} \qquad \sigma \equiv \frac{J}{\mathcal{E}}$$

- The ***conductivity*** of a semiconductor is $\boxed{\sigma \equiv qp\mu_p + qn\mu_n}$
  - Unit: mho/cm

- The ***resistivity*** of a semiconductor is $\boxed{\rho \equiv \frac{1}{\sigma}}$
  - Unit: ohm-cm

Resistivity Dependence on Doping



For n-type material:

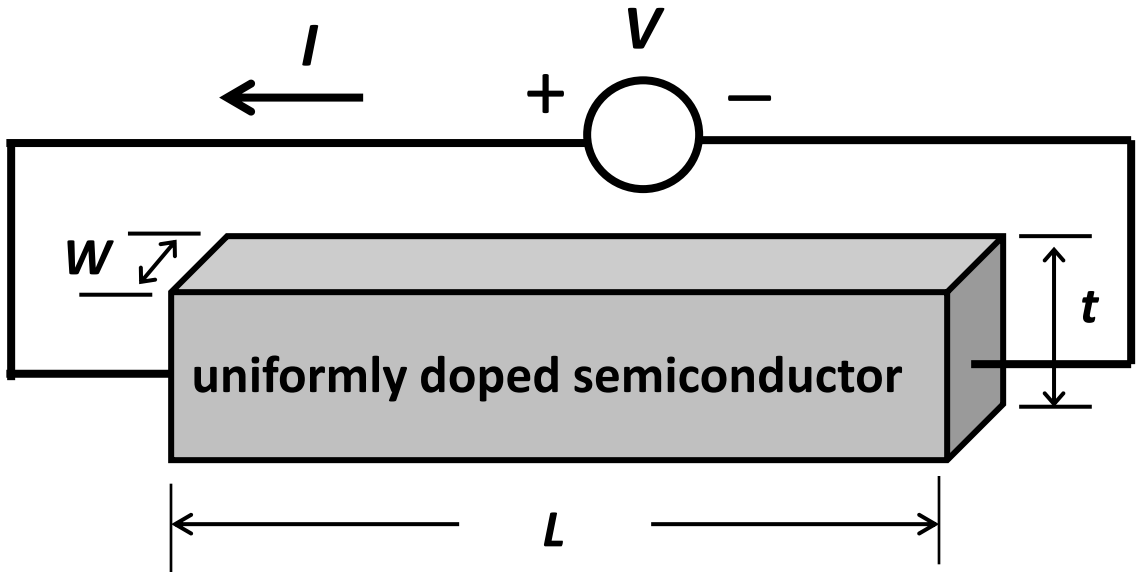
$$\rho \cong \frac{1}{qn\mu_n}$$

For p-type material:

$$\rho \cong \frac{1}{qp\mu_p}$$

Note: This plot (for Si) does not apply to compensated material (doped with both acceptors and donors).

Electrical Resistance



Resistance

$$R \equiv \frac{V}{I} = \rho \frac{L}{Wt}$$

[Unit: ohms]

where ρ is the resistivity

Example: Resistivity Calculation

What is the resistivity of a Si sample doped with $10^{16}/cm^3$ Boron?

Answer:

$$N_A = 10^{16}\ \text{cm}^{-3}, N_A = 0$$
$$p = 10^{16}\ \text{cm}^{-3},\ n = 10^{4}\ \text{cm}^{-3}$$

$$np = n_i^2 = 10^{20}\ \text{cm}^{-6}$$

$$\rho = \frac{1}{qn\mu_n + qp\mu_p} \cong \frac{1}{qp\mu_p}$$

$$= \left[(1.6\times10^{-19})(10^{16})(450)\right]^{-1} = 1.4\,\Omega - \text{cm}$$

Example: Compensated Doping

Consider the same Si sample doped with $10^{16}/cm^3$ Boron, and *additionally* doped with $10^{17}/cm^3$ Arsenic. What is its resistivity?

Answer:

$$N_D = 10^{17}\ \text{cm}^{-3}, N_D = 10^{16}\ \text{cm}^{-3}$$
$$n = 9 \times 10^{16}\ \text{cm}^{-3}$$

$$\rho = \frac{1}{qn\mu_n + qp\mu_p} \cong \frac{1}{qn\mu_n}$$

$$= \left[(1.6\times10^{-19})(9\times10^{16})(750)\right]^{-1} = 0.93\,\Omega - \text{cm}$$

Example: *T* Dependence of ρ

Consider a Si sample doped with 10^{17} As atoms/cm^3. How will its resistivity change when *T* is increased from 300K to 400K?

Answer:

The temperature dependent factor in σ (and therefore ρ) is μ_n.

From the mobility *vs.* temperature curve for 10^{17} cm^{-3}, we find that μ_n decreases from 770 at 300K to 400 at 400K.

Thus, ρ increases by $\frac{770}{400} = 1.93$

Summary

- **Thermal equilibrium:**
  - Balance between internal processes with no external stimulus (no electric field, no light, *etc*.)
  - **Fermi function** $$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$
    - Probability that a state at energy *E* is filled with an electron, under equilibrium conditions.
    - **Boltzmann approximation:**

      For high *E*, *i.e.* $E - E_F > 3kT$: $f(E) \cong e^{-(E - E_F)/kT}$

      For low *E*, *i.e.* $E_F - E > 3kT$: $1 - f(E) \cong e^{-(E_F - E)/kT}$

Summary (cont'd)

- **Relationship between E_F and n, p :**

$$n = N_c e^{-(E_c - E_F)/kT} = n_i e^{(E_F - E_i)/kT}$$

$$p = N_v e^{-(E_F - E_v)/kT} = n_i e^{(E_i - E_F)/kT}$$

- **Intrinsic carrier concentration :**

$$n_i = \sqrt{N_c N_v}\, e^{-E_G/2kT}$$

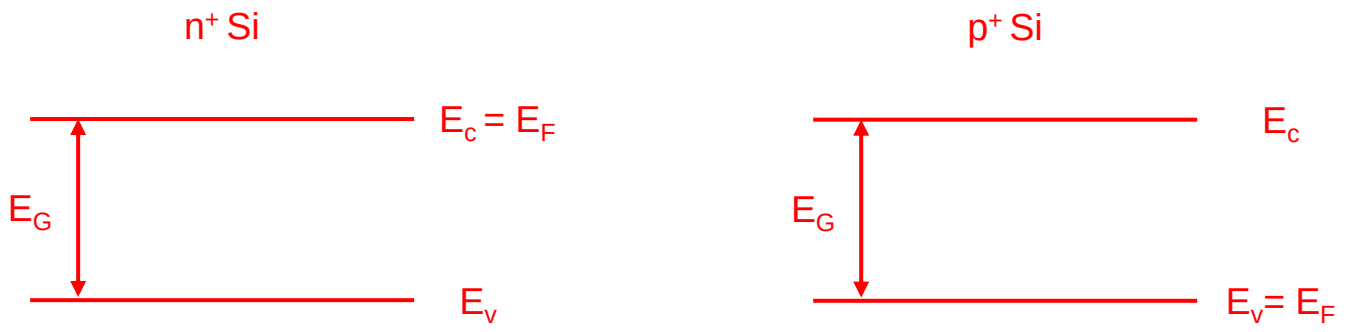
- **The intrinsic Fermi level, E_i, is located near midgap.**

Summary (cont'd)

- If the dopant concentration exceeds 10^{18} cm^{-3}, silicon is said to be degenerately doped.
  - The simple formulas relating *n* and *p* exponentially to E_F are not valid in this case.

For degenerately doped n-type (n+) Si: $E_F \cong E_c$

For degenerately doped p-type (p+) Si: $E_F \cong E_v$



Summary

- Electrons and holes can be considered as quasi-classical particles with effective mass *m**
- In the presence of an electric field $\mathcal{E}$, carriers move with **average drift velocity $v_d = \mu\mathcal{E}$, μ is the carrier mobility**
  - **Mobility decreases w/ increasing <u>total concentration of ionized dopants</u>**
  - **Mobility is dependent on temperature**
    - decreases w/ increasing *T* if lattice scattering is dominant
    - decreases w/ decreasing *T* if impurity scattering is dominant
- The conductivity (σ) hence the resistivity (ρ) of a semiconductor is dependent on its mobile charge carrier concentrations and mobilities

$$\sigma \equiv qp\mu_p + qn\mu_n \qquad \rho \equiv 1/\sigma$$

Acknowledgement

- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
- Professor Tsu-Jae King Liu EE130/230A notes